

Deepak Mallampati

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Kent, OH -- F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) shipping production agentic GenAI and RAG systems in microservices. Track record: RAG (~35% retrieval precision gain), agentic LLM workflows with grounding controls (>30% hallucination reduction), and structured prompt transformation layers (+30% contextual alignment in multimodal pipelines). Strong production discipline: FastAPI/Docker microservices, structured logging, evaluation pipelines, and OAuth-gated user flows. Independently shipped **Manga Lens** (live Chrome extension, 4 AI providers across 29 sites) and **Agentic Healthcare Claims** with schema-validated multi-agent contracts. Comfortable across LangChain, LangGraph-style orchestration, fine-tuning, function calling, and vector search.

CORE COMPETENCIES

Agentic AI & Multi-Agent Orchestration . GenAI in Production . LangChain / LangGraph / LlamaIndex . RAG & Vector Search . Prompt Engineering & Fine-Tuning . Tool Use & Function Calling . Python / TypeScript / FastAPI . Production Microservices . OAuth & Token-Based Access . Evaluation & Observability . Technical Design Specs

EXPERIENCE

Progress Solutions Inc.

Jul 2025 -- Present

Jr. AI/ML Engineer Trainee -- USA | AI Consulting, Production LLM Systems

- Developed **agentic LLM workflows** with structured reasoning, tool discipline, grounding rules, and inter-step evaluation; improved agent stability ~25% and reduced hallucination rate >30%.
- Built and deployed **RAG systems** with recursive semantic chunking + transformer embeddings; ~35% retrieval precision gain; grounded-response alignment exceeded 90%.
- Packaged ML/LLM inference as **FastAPI** microservices containerized with Docker; structured logging + load simulation reduced post-deployment defects by ~30%.
- Built technical design docs + system-limitation specs for stakeholders; lineage + audit trails for regulated environments.
- Shipped predictive ML pipelines (scikit-learn, XGBoost) for risk scoring; improved recall 15-20% on high-risk categories.

Suvidha Foundation

Jan 2022 -- Mar 2022

Machine Learning Engineer -- Hyderabad, India | Applied NLP

- Built transformer-based video summarization for 5,000+ sessions; reduced manual review time 60-70%.
- Implemented document Q&A with RAG; reduced hallucinated outputs ~30%.

Energy Solutions International (Emerson)

Jun 2022 -- Apr 2023

Database & DevOps Performance Engineer (Intern) -- Hyderabad, India

- Built CI/CD pipelines with Jenkins; reduced deployment errors >30%; improved release cycle ~35-40%.
- Designed Grafana dashboards for production SQL systems; cut incident recurrence ~25%.

PROJECTS

Manga Lens shipped -- Chrome Web Store

Solo-built Chrome extension integrating **4 AI vision providers** with provider-agnostic abstraction. Manifest V3 + content scripts + service workers; multi-section capture pipeline; 7-day translation cache; 29 site selector configs; scoped host permissions and privacy policy. Production discipline for consumer-scale AI features.

TypeScript, Manifest V3, multi-provider abstraction, Chrome APIs

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline (Intake -> Validation -> Consistency -> Duplicate Detection -> Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation; ANN similarity for duplicates; explainable risk scoring with audit-ready reasoning traces.

Python, LangChain, vector search, FastAPI, Docker

Dream Decoder -- AI Multimodal Creative Platform

Full-stack FastAPI + React/TS app with coordinated multimodal APIs. **Intermediate structured prompt transformation layers** improved contextual alignment ~**30%** and first-pass success ~**25-30%**.

FastAPI, React, TypeScript, Perplexity Sonar, Replicate

Agentic Pixel Character Synthesis Engine

Phased GenAI system with Stable Diffusion fine-tuning + LoRA for identity consistency; ControlNet + OpenPose/MediaPipe pose conditioning; **LLM-based agent orchestrator** that decomposes high-level prompts into generation tasks.

PyTorch, HF Diffusers, FastAPI, Docker, LoRA, ControlNet

EDUCATION

Master's Degree -- *Kent State University, Ohio, USA*

Aug 2023 -- May 2025

TECHNICAL SKILLS

Languages & Frameworks: Python, TypeScript, JavaScript, React, FastAPI, Flask, SQL (PostgreSQL, T-SQL)

AI/LLM & Agentic: LangChain, LangGraph (familiar), LlamaIndex, RAG, multi-agent orchestration, prompt engineering, fine-tuning, function calling, structured outputs, evaluation pipelines, guardrails

Cloud / MLOps: Docker, AWS (familiar), Databricks/SageMaker/Vertex AI (familiar), CI/CD (Jenkins), Grafana, REST APIs, OAuth, MLOps/LLMOps

ML Libraries: PyTorch, HF Transformers, scikit-learn, XGBoost, Diffusers, ControlNet, LoRA

Work Authorization: F-1 OPT -- authorized to work in the United States now; future sponsorship required